Long-Term Impacts of Early-Life Cash Transfers: A Causal Forest Approach

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Abstract. This paper examines the long-term effects of early-life cash transfers on longevity using historical data from the U.S. Mothers' Pension program. While standard approaches focus primarily on average effects, this study applies Generalized Random Forests to estimate Conditional Average Treatment Effects and explore treatment heterogeneity. Using a doubly robust, orthogonalized machine learning framework, the preferred specification suggests that program acceptance increased child lifespan by approximately 1.55 years on average. The results also provide evidence of meaningful heterogeneity, indicating that estimated treatment effects vary with local characteristics such as baseline educational expenditures.

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1 Introduction

Cash transfers are a cornerstone of modern social policy, yet debates regarding their long-term effectiveness and optimal design remain unresolved. While empirical evaluations frequently focus on average treatment effects (ATEs), such aggregate measures can obscure substantial variation in how different subgroups respond to financial assistance. Understanding this heterogeneity is crucial for designing more effective, equitable, and targeted interventions. To that end, this paper investigates the long-term effects of early-life cash transfers on longevity, with a specific focus on uncovering the drivers of treatment effect heterogeneity across individuals and regions.

To explore this question, I apply Causal Forests—a state-of-the-art causal machine learning method—to historical administrative data from the U.S. Mothers' Pension (MP) program. As one of the earliest forms of public cash assistance provided to poor families, the MP program offers a uniquely valuable natural experiment for studying the lifetime impacts of welfare. By leveraging this rich archival dataset, I estimate both average and Conditional Average Treatment Effects (CATEs), identifying how the long-run impacts of childhood financial support vary according to baseline family, county, and policy characteristics.

The remainder of the paper is structured as follows. Section 2 reviews the existing literature, highlighting the original MP program study by Aizer et al. (2016) alongside recent methodological advances in causal machine learning. Section 3 outlines the empirical identification strategy and details the Causal Forest econometric framework. Section 4 describes the data construction and variables. Section 5 presents the results, including estimates of the average treatment effect, formal tests for heterogeneity, and various robustness checks. Finally, Section 6 concludes with a discussion of the findings and directions for future research.

2 Literature Review

2.1 Research on Early-Life Poverty and Cash Transfers

A large body of research highlights the long-term adverse consequences of early-life poverty on children's outcomes, including education, labor market performance, and ultimately, mortality (Almond and Currie, 2011). In the United States, over 20 percent of children

were living in poverty as recently as 2010, making the question of how to mitigate these effects highly relevant for policy. Parental income has consistently been identified as a strong predictor of children's educational attainment (Barrow and Schanzenbach, 2012; Reardon, 2011) and adult health outcomes (Case et al., 2002). However, whether direct cash transfers to poor families translate into long-term benefits for children remains an open question (Currie, 1998).

There are several reasons why means-tested cash transfers may not always deliver positive outcomes. The amounts disbursed may be too small to substantially affect child well-being, or parents may not allocate funds in ways that benefit their children due to lack of information or other constraints (Dizon-Ross, 2014). In addition, transfers may induce behavioral changes among parents, such as altering labor supply, fertility decisions, or marriage patterns, that could have unintended consequences for children's development (Moffitt, 1998).

Methodologically, studying the lifetime impact of welfare transfers presents significant challenges. It is often difficult to construct a credible counterfactual, that is, what would have happened to children in the absence of such programs. Furthermore, suitable long-term data on recipients and comparison groups are rare (Aizer et al., 2016). Prominent longitudinal datasets like the NLSY and PSID include relatively few welfare recipients and suffer from attrition. Administrative records for earlier welfare programs such as Aid to Dependent Children (ADC) are largely incomplete or unavailable. More recent cohorts complicate evaluation because recipients typically participate in multiple overlapping programs, such as Medicaid or food stamps, which makes it difficult to isolate the effects of cash transfers alone (Aizer et al., 2016).

2.2 The Original Study (Aizer et al., 2016)

Aizer et al. (2016) provide one of the most comprehensive analyses of the long-term effects of early-life cash transfers on children's outcomes. Their study evaluates the U.S. Mothers' Pension (MP) program (1911–1935), the first government-sponsored welfare program aimed at supporting poor families with children. Using a rich historical dataset, the authors track individual-level outcomes of over 16,000 boys whose mothers applied for MP benefits, linking these records to U.S. Census data, WWII enlistment records, and death certificates. The key identification strategy compares children of accepted applicants to those of similarly poor rejected applicants, leveraging administrative decisions and ensuring that both groups faced

comparable conditions at baseline.

Their findings show significant long-run impacts: boys whose families received MP support lived, on average, one year longer, obtained higher levels of education (+0.34 years), had lower incidence of underweight status, and earned higher incomes in adulthood. The effects were particularly pronounced among the poorest families. Mechanistically, the authors attribute most of the longevity gains to improvements in nutrition, education, and income.

The study highlights the potential of well-targeted cash transfers to improve life trajectories, while also acknowledging important limitations: inability to track women's outcomes, exclusion of African-Americans due to data limitations, and residual concerns about unobservable selection bias. Notably, while the paper explores subgroup effects (by income and urban residence), its methodological approach relies on average treatment effects and basic subgroup comparisons. As such, it does not fully exploit modern tools for assessing treatment effect heterogeneity — a gap that this paper seeks to address through the application of Causal Forests.

2.3 Causal Machine Learning and Its Relevance for This Study

Recent advances in causal machine learning (CML) offer new opportunities for evaluating the heterogeneity of treatment effects in social policy interventions. In particular, causal forests (Athey and Wager, 2018; Athey et al., 2019) have emerged as a powerful tool for identifying CATEs across different subgroups in observational data. Unlike traditional regression-based methods that primarily focus on estimating ATEs, causal forests are designed to detect variation in treatment effects across individuals and contexts.

This feature is especially relevant for the current study of early-life cash transfers and their long-term impact on longevity. Prior research on social welfare programs has typically reported average effects (Aizer et al., 2016), which may conceal important heterogeneity. For example, children from different family backgrounds, regions, or maternal marital status may experience varying long-term benefits from cash transfers. Causal forests enable us to systematically explore such heterogeneity without the need to pre-specify subgroups or interaction terms.

Methodologically, causal forests address several key challenges inherent in historical welfare data. First, they correct for potential confounding by using estimated propensity scores and orthogonalization techniques, improving robustness in non-experimental settings.

Second, they account for clustered errors, such as school or regional effects, through cluster-robust estimation. Third, by applying sample-splitting (honest estimation), causal forests help mitigate overfitting, providing more credible estimates of heterogeneous treatment effects (Athey and Wager, 2019).

For this research question, whether early-life cash transfers affect longevity and how these effects vary across subpopulations, causal forests are particularly well-suited. They allow us to move beyond average effects and uncover patterns of heterogeneity that can inform more targeted and effective social policies.

3 Empirical Strategy

3.1 Identification Strategy

The main challenge in estimating the causal impact of early-life cash transfers on long-term outcomes such as longevity is constructing a credible counterfactual—namely, what the outcomes of treated children would have been in the absence of receiving transfers. In my study, I follow the approach of Aizer et al. (2016) and leverage quasi-experimental variation arising from the administrative process of the Mothers' Pension program. Specifically, I compare outcomes for children whose mothers applied for the MP program and were ultimately **accepted**, to outcomes for those whose mothers were **rejected** upon further investigation.

This strategy relies on the assumption that accepted and rejected applicants are, conditional on observables, similar in their underlying characteristics. In practice, all applicants were initially deemed eligible—i.e., poor mothers with dependent children—and the decision to grant or deny transfers was based on caseworker evaluations that frequently hinged on small differences in resources or local implementation practices rather than on large pre-existing differences in socioeconomic status or child endowments. Aizer et al. (2016) document that rejected mothers were on average slightly better off but broadly comparable to accepted mothers across observable measures such as family size, marital status, and predicted income.

Formally, identification relies on the *conditional unconfoundedness assumption*:

$$\{Y_i(1), Y_i(0)\} \perp MP_i \,\big|\, X_i, \tag{1}$$

where:

- $Y_i(1)$ and $Y_i(0)$ are the potential outcomes (log age at death) if treated and untreated,
- MP_i is an indicator equal to 1 if the mother's pension application was accepted,
- X_i denotes a rich vector of pre-treatment covariates.

This assumption implies that, *conditional on observed characteristics*, the assignment of treatment approximates random variation. Several features of the Mothers' Pension program and the accompanying dataset enhance the plausibility of this assumption. First, the administrative records provide extensive baseline information—including detailed measures of family structure, maternal demographics, economic circumstances, and application timing—which allows for rigorous control of observable differences. Furthermore, the administrative process itself introduced a degree of quasi-random variation. Eligibility was determined through local caseworker assessments that often hinged on subjective judgments and fluctuating local budgets. In many states, severe funding constraints forced administrators to deny applications from otherwise eligible families, generating variation that is plausibly orthogonal to unobservable determinants of long-run health. Additionally, because the Mothers' Pension was the primary form of public support available to poor families during this period, the potential for confounding from other contemporaneous welfare interventions is substantially limited.

Despite these institutional advantages, potential threats to identification remain. Primary among these is the risk of residual selection on unobservables, as more motivated or better-networked mothers might have been more successful in securing support. There is also the potential for differential attrition if families receiving transfers were more likely to remain in the state, thereby systematically affecting long-term record linkage. Finally, incomplete administrative records introduce the possibility of measurement error, potentially misclassifying a family's true treatment status.

To mitigate these concerns, the empirical analysis incorporates several robust strategies. I include a rich array of baseline covariates, alongside state and cohort fixed effects, to systematically absorb biases stemming from observed and partially observed determinants of treatment. To account for intra-county correlation in unobserved shocks and localized administrative practices, standard errors are clustered at the county level. Moreover, the Causal Forest framework addresses confounding directly through orthogonalization, which residualizes both the longevity outcomes and the treatment propensity via preliminary regression forests. Finally, to ensure the reliability of the findings, the main estimations are subjected to

extensive sensitivity analyses—including tests with alternative variable subsets and sample restrictions—to formally assess the robustness of the estimates (Rosenbaum, 2002).

The main empirical specification is an accelerated failure time (AFT) model for log age at death:

$$\log\big(\text{Age at Death}_i\big) = \alpha + \beta \cdot MP_i + \gamma' X_i + \delta_s + \lambda_c + \eta_t + \varepsilon_i, \tag{2}$$

where δ_s, λ_c, and η_t denote state, county, and cohort fixed effects respectively. The coefficient β measures the average treatment effect on log longevity.

Under the maintained assumption that, conditional on observables and fixed effects, treatment assignment is as good as random among applicants, β recovers the causal impact of the Mothers' Pension cash transfer.

This approach has several strengths relative to alternative designs. First, it leverages administrative records rather than self-reported survey data, reducing measurement error and attrition. Second, the design isolates the effect of cash transfers without the confounding influence of contemporaneous welfare programs (since, during the period studied, the MP was the primary form of public assistance available). Finally, the comparison of accepted and rejected applicants represents a conservative strategy, as rejected families often had slightly higher pre-application incomes. This suggests that my estimates may be interpreted as a **lower bound** on the true treatment effect.

3.2 Econometric Method

The econometric approach in this study is based on *Causal Forests*, a machine learning method designed to estimate conditional average treatment effects in observational data. This technique extends the well-known Random Forest algorithm by explicitly targeting heterogeneity in treatment effects, rather than focusing solely on prediction of outcomes (Athey and Wager, 2019).

Formally, let Y_i denote the outcome of interest (the log of age at death), and W_i a binary treatment indicator equal to 1 if the mother's pension application was accepted and 0 otherwise. For each unit i, define potential outcomes:

$$Y_i(1) = \text{log(age at death) if treated,}$$

$$Y_i(0) = \text{log(age at death) if untreated.}$$

The quantity of interest is the Conditional Average Treatment Effect:

$$\tau(x) = \mathbb{E}\big[Y_i(1) - Y_i(0) \mid X_i = x\big],$$

where X_i denotes a high-dimensional vector of observed covariates describing the child, family, cohort, and region.

Estimating $\tau(x)$ in historical, non-experimental data requires addressing two main challenges: Selection Bias and High Dimensionality. Causal Forests address these challenges through several innovations.

Orthogonalization and R-Learner. First, the algorithm residualizes both the outcome and treatment. Specifically, it estimates:

$$m(x) = \mathbb{E}[Y_i \mid X_i = x],$$

$$e(x) = \mathbb{E}[W_i \mid X_i = x].$$

These are the conditional expectations (the predicted outcome and the propensity score), estimated via separate regression forests. The residuals are computed as:

$$\tilde{Y}_i = Y_i - m(X_i),$$

$$\tilde{W}_i = W_i - e(X_i).$$

By focusing on the relationship between these residuals, the forest achieves orthogonalization, reducing bias from confounding variables (Robinson, 1988).

The estimation of the **Conditional Average Treatment Effect** function $\tau(x)$ is based on the *R-learner* objective proposed by Nie and Wager (2017). Specifically, $\tau(x)$ is obtained by minimizing the following penalized loss function:

$$\hat{\tau}(\cdot) = \arg\min_{\tau} \sum_{i=1}^{n} \Big[\big(Y_i - \hat{m}^{(-i)}(X_i) - \tau(X_i)\,(W_i - \hat{e}^{(-i)}(X_i))\big)^2\Big] \;+\; \Lambda_n(\tau).$$

Here, $\hat{m}^{(-i)}(X_i)$ and $\hat{e}^{(-i)}(X_i)$ are out-of-bag predictions of the outcome and propensity

score, respectively, and $\Lambda_n(\tau)$ is a regularization term that controls the complexity of the estimated function.

This approach has the advantage that if the true treatment effect function $\tau(x)$ is simpler than the main effect or propensity functions, it can be estimated more efficiently and with faster convergence.

Honest Estimation and Sample Splitting. To avoid overfitting, Causal Forests use *honest estimation*: the sample is split into two parts. One part is used to grow the trees (selecting the splits), while the other is used to estimate treatment effects within the leaves. This ensures that the same data is not used both to decide where to split and to estimate effects, improving the validity of inference.

Tree Growing and Splitting. Causal Forests construct an ensemble of decision trees grown on random subsamples of the data. Each tree partitions the covariate space recursively by selecting splits that *maximize the heterogeneity in estimated treatment effects* across child nodes.

Average Treatment Effect. Causal forests can also be interpreted as *adaptive kernel methods*, where each observation contributes to predictions with a weight depending on how often it shares leaves with the target point across all trees. Specifically, the weights $\alpha_i(x)$ capture how frequently observation i appears in the same leaf as x.

Using these weights, the *average treatment effect* can be estimated by the following formula:

$$\hat{\tau} = \frac{\sum_{i=1}^{n} \alpha_i(x) \left(Y_i - \hat{m}^{(-i)}(X_i)\right) \left(W_i - \hat{e}^{(-i)}(X_i)\right)}{\sum_{i=1}^{n} \alpha_i(x) \left(W_i - \hat{e}^{(-i)}(X_i)\right)^2}.$$

This approach intuitively means that each observation's contribution to the estimate depends on how similar it is to the target point in the forest partitioning, which allows for flexible, data-driven estimation of treatment effects without strong parametric assumptions.

Implementation Details. The estimation proceeds as follows:

1. **Covariates**:

    - The model includes state fixed effects, cohort indicators, and detailed baseline characteristics of the child and household.

2. **Variable Selection**:
    - An initial causal forest is fitted to compute variable importance scores.
    - Variables with above-mean importance are retained.
3. **Final Causal Forest**:
    - The final model is trained on the selected variables, with:
        - *Orthogonalization* (using pre-fitted regression forests),
        - *Honest estimation*,
        - *Clustering* at the county level to account for within-county dependence,
        - *Automatic tuning* of hyperparameters (`tune.parameters = "all"`).
4. **Inference**:
    - Pointwise standard errors for CATEs are derived from subsample variability across trees.
    - ATE confidence intervals are computed using influence-function–based variance estimates.

This flexible machine learning approach allows for rich, nonlinear treatment effect heterogeneity while maintaining valid inference under the assumption of unconfoundedness conditional on observed covariates.

4 Data Description

4.1 Data Sources and Sample Construction

This study builds on the historical dataset from the Mothers' Pension program operating from 1911 to 1935. The original data collection effort, as documented by Aizer et al. (2016), assembled detailed administrative records from eleven states: Idaho, Illinois, Iowa, Minnesota, Montana, North Dakota, Ohio, Oklahoma, Oregon, Washington, and Wisconsin. Although approximately 80,000 children were initially recorded in the MP archives, this study focuses on the subset of observations that meet stringent criteria for identification and outcome measurement.

Following sample restrictions similar to Aizer et al. (2016), the analysis is limited to male children born between 1900 and 1925, for whom reliable mortality outcomes can be obtained. Female children are excluded due to difficulties in tracing their later life outcomes because of name changes after marriage. Furthermore, only counties with data on both accepted and rejected applicants are retained to allow for credible causal inference.

Mortality outcomes, which are central to this study's focus on longevity, are derived by linking individuals in the MP records to the Social Security Death Master File (DMF), covering deaths reported between 1965 and 2012. The analysis uses exact name and birth-date matches to ensure high-quality linkages. To further strengthen the reliability of the outcome variable, only individuals with a unique match to the DMF are included in the final analytic sample. After applying these restrictions, the working dataset consists of 7,800 uniquely matched male children, drawn from an original pool of around 16,000 applicants with complete and usable information.

The dataset is enriched with a wide set of covariates relevant for understanding heterogeneity in treatment effects. These include individual-level child and family characteristics, such as family size, maternal marital status, and ages of siblings; county-level socioeconomic indicators; and state-level time-varying policy measures (e.g., generosity of MP benefits, education laws, and social spending). These variables allow the analysis to adjust for potential confounding factors and to explore variation in treatment effects across subgroups using causal forest methods. Summary statistics for the estimation sample (Matched Data) are presented in Appendix, Table 3.

Table 1 reports the associations between child and family characteristics and the likelihood of acceptance into the Mothers' Pension program, as well as the generosity of benefits (logged amount and duration), based on the matched sample. Several systematic patterns emerge. First, family size is positively associated with both the probability of acceptance and the level of benefits: larger families were more likely to receive support, and in greater amounts and duration. This finding aligns with qualitative evidence that courts prioritized families with greater financial need. Second, younger children were associated with higher acceptance rates and larger transfers, while families with older children—closer to working age—were less likely to receive support, consistent with program targeting toward families with dependent children. Third, certain marital status variables, such as widowhood (omitted category), were more predictive of acceptance and generosity than other statuses like divorce or husband

absence, again consistent with the program's original design to support widows. Finally, the presence of missing birth information is correlated with longer transfer durations, which may reflect unobserved characteristics such as lower literacy among recipients, as discussed in Aizer et al. (2016).

Overall, these patterns suggest that while accepted and rejected applicants were similar in many observable characteristics—supporting the use of rejected applicants as a counterfactual group—there are small but systematic differences: accepted families were generally larger, had younger children, and likely faced greater financial need. These differences underscore the importance of controlling for observable confounding when estimating causal effects of the MP program.

4.2 Data Preparation and Variable Definition

The first step in the analysis involved loading and preparing the matched dataset. The outcome variable is the logarithm of age at death based on the MP birth date, defined as:

$$Y_i = \log(\text{age at death}_i) \tag{3}$$

The treatment variable is a binary indicator of whether the MP application was accepted:

$$W_i = \mathbb{1}\{\text{MP application accepted}_i\} \tag{4}$$

To estimate the conditional average treatment effect, a rich set of pre-treatment covariates was used. These are denoted by X_i, and include the following groups of variables:

- **Mother characteristics**: divorced status, husband's absence, and missing marital status indicator.
- **Child characteristics**: child's age, length of family name, dummies for sibling count (2 to 8), minimum and maximum child age in household.
- **County characteristics (1910 Census)**: average socio-economic index (SEI), standard deviation of SEI, share urban, share widowed, share of children in single-mother households, poverty rate, female labor force participation, child labor, and average value of farmland.

Table 1: Determinants of Acceptance and Generosity of Transfers (Matched Sample)

	Accepted	**ln(amount)**	**ln(duration)**
Child age (years)	-0.00277	0.00177	-0.0139
	[0.004]	[0.007]	[0.017]
Length of family name	-0.000319	0.00240	0.00779
	[0.003]	[0.003]	[0.010]
Family size = 2	0.0349**	0.273***	0.331***
	[0.017]	[0.033]	[0.061]
Family size = 3	0.0511***	0.491***	0.511***
	[0.019]	[0.042]	[0.062]
Family size = 4	0.0624***	0.624***	0.545***
	[0.024]	[0.055]	[0.061]
Family size = 5	0.0570*	0.755***	0.578***
	[0.032]	[0.061]	[0.092]
Family size = 6	0.0626**	0.807***	0.588***
	[0.030]	[0.078]	[0.120]
Family size = 7	0.0654*	0.922***	0.573***
	[0.038]	[0.093]	[0.120]
Family size = 8+	0.0704	0.975***	0.586***
	[0.046]	[0.077]	[0.125]
Age of oldest kid in family	-0.00129	-0.00547*	-0.00665
	[0.002]	[0.003]	[0.006]
Age of youngest kid in family	-0.00258	-0.0133***	-0.0178**
	[0.002]	[0.003]	[0.007]
Divorced	-0.0415	0.00865	0.0148
	[0.034]	[0.028]	[0.030]
Husband abandoned/prison/hospital	-0.00448	-0.00651	-0.0815***
	[0.010]	[0.017]	[0.026]
Mother's marital status unknown	-0.116	-0.0333	0.0764
	[0.076]	[0.038]	[0.065]
Day or month of birth missing	0.0399	0.0433	0.356*
	[0.039]	[0.065]	[0.211]
Observations	7,859	6,820	3,677
Mean	0.875	5.508	1.340

Robust standard errors in brackets.

*** Significant at the 1 percent level.

** Significant at the 5 percent level.

* Significant at the 10 percent level.

Source: Aizer et al. (2016).

- **County fixed effects**: dummy variables for counties (CID2 to CID75).
- **Matching variables**: indicator for missing birth date.
- **State fixed effects**: dummy variables for states (S2 to S11).
- **State-year characteristics**: variables reflecting manufacturing wages, educational laws, and state-level public spending on welfare, schools, and charity.
- **Cohort variables**: year of birth (linear and squared).
- **Cohort fixed effects**: birth-year dummy variables (BY2 to BY26).

Following the original paper by Aizer et al. (2016), four different specifications were implemented for robustness checks:

1. **Baseline model (state and cohort fixed effects only)**:

$$X_i = \text{State dummies} + \text{Cohort dummies}$$

2. **Extended model (all controls, no county fixed effects)**:

$$\begin{aligned} X_i = \text{Mother} + \text{Child} + \text{Matching} + \text{County characteristics} \\ + \text{State-year characteristics} + \text{Cohort dummies} \end{aligned} \tag{5}$$

3. **Full model (with county fixed effects)**:

$$\begin{aligned} X_i = \text{Mother} + \text{Child} + \text{Matching} + \text{County dummies} \\ + \text{State-year characteristics} + \text{Cohort dummies} \end{aligned} \tag{6}$$

4. **Alternative outcome (log age at death using SSA birth date)**:

$$Y_i = \log(\text{age at death}_{\text{SSA DOB}})$$

with the same covariates as specification (3).

5 Results

5.1 Average Treatment Effects

5.1.1 Estimating the Propensity Score

To estimate the likelihood of being accepted into the MP program conditional on observed covariates, I compute *propensity scores* using a logistic regression model. This step is essential for the subsequent estimation of treatment effects using causal forests, which rely on the overlap and balance between treated and untreated observations.

The model takes the form:

$$\Pr(W_i = 1 \mid X_i) = \Lambda(X_i\beta) \tag{7}$$

where W_i is a binary indicator equal to 1 if the application was accepted and 0 otherwise, and X_i denotes a rich set of pre-treatment covariates, as discussed and specified in the previous subsubsection.

The logistic regression model is estimated using only complete cases (i.e., observations with no missing values in any covariate). The predicted propensity scores $\hat{e}(X_i)$ are then stored for each observation.

Summary statistics indicate that the estimated scores are generally high:

- **Min**: 0.11
- **Median**: 0.91
- **Mean**: 0.87
- **Max**: 1.00

Figure 1 illustrates the distribution of propensity scores for both treated and untreated individuals. The treated group ($W = 1$) is highly concentrated at the upper end of the distribution, while the untreated group ($W = 0$) occupies a smaller and flatter region. This indicates limited overlap between the two groups, which poses a challenge for causal inference.

The dataset also exhibits a treatment imbalance:

- **Accepted**: 6,877 observations ($\approx$ 87.5%)

- **Rejected**: 983 observations ($\approx 12.5\%$)

Such imbalance results in a relatively narrow region of common support. To mitigate this issue, trimming observations with extreme propensity score values may be considered.

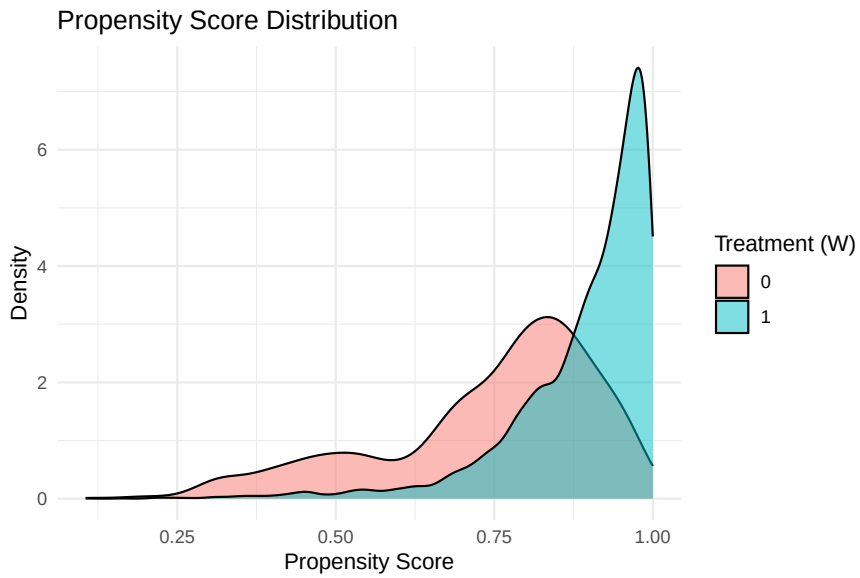

Figure 1: Propensity Score Distribution. Source: data from Aizer et al. (2016); author's own calculations.

5.1.2 Estimating the ATE Using Causal Forests

Next, I estimate the ATE using the causal forest method. I begin by growing an initial forest and augmenting it with additional trees, following the standard procedure in the `grf` package.

The results for four specifications are as follows:

1. **Baseline Model (state and cohort fixed effects):**

   $\widehat{\text{ATE}} = 0.012$ (95% CI: ± 0.014)

   $\widehat{\text{ATE}}_{\text{control sample}} = 0.016$ (95% CI: ± 0.014).

2. **Extended Model (all controls, no county fixed effects):**

   $\widehat{\text{ATE}}$: 0.016 (95% CI: ± 0.014)

   $\widehat{\text{ATE}}_{\text{control sample}}$: 0.015 (95% CI: ± 0.014)

3. **Full Model (with county fixed effects):**

   $\widehat{\text{ATE}}$: 0.015 (95% CI: ± 0.013)

   $\widehat{\text{ATE}}_{\text{control sample}}$: 0.015 (95% CI: ± 0.014)

   Implied increase in lifespan: **1.09 years** (95% CI: 0.1 to 2.09 years)

4. **Alternative outcome (SSA date of birth):**

   $\widehat{\text{ATE}}$: 0.014 (95% CI: ± 0.013)

   $\widehat{\text{ATE}}_{\text{control sample}}$: 0.014 (95% CI: ± 0.014)

All models are estimated with `equalize.cluster.weights = FALSE`, since enabling this option—by giving equal weight to each cluster rather than each observation—reduces variation across trees and results in degenerate variable importance measures (i.e., all zeros), making it impossible to assess which features drive treatment effects. While the treatment effect estimates are consistently positive, they are smaller than those reported in the original MP study ($\widehat{\text{ATE}} \approx 0.0182$, significant at 5%, corresponding to a lifespan gain of 1.1–1.3 years).

Moreover, several specifications yield confidence intervals that include zero, indicating limited statistical significance. To address potential violations of the overlap assumption, I proceed by trimming observations with extreme propensity scores. This step is motivated by the limited region of common support, as illustrated in Figure 1.

5.1.3 Trimming Based on Propensity Score

To improve the overlap between treated and control observations, I apply *trimming*, excluding units with extreme estimated propensity scores that fall outside the region of common support. This is necessary due to the imbalance in treatment probabilities observed earlier.

Based on the propensity score distribution and quantiles, I set the trimming bounds as follows: a lower bound of 0.65 and an upper bound of 0.97. These thresholds capture the central region where treated and control units overlap more substantially.

As a result, 2,317 observations were dropped from the estimation sample, and 5,542 units were retained for further analysis.

Figure 2 displays the kernel density estimates of the propensity scores for treated ($W = 1$) and control ($W = 0$) groups, separated by whether the observation was retained after trimming. The panel labeled `TRUE` on the right shows the distribution for the trimmed (retained) sample, where the region of overlap is more pronounced. This confirms the appropriateness of the trimming bounds.

Figure 2: Propensity Score Distribution (Full vs Trimmed Sample). Source: data from Aizer et al. (2016); author's own calculations.

5.1.4 Causal Forest Estimation on Trimmed Data

After applying propensity score trimming, I re-estimated the causal forest on the resulting trimmed sample.

Model Choice. The forest model estimated on the trimmed sample, using the full set of controls and including county fixed effects, serves as the **main specification** for our analysis.

Average Treatment Effect. The estimation follows the standard two-step approach proposed by Wager and Athey (2019). The estimated average treatment effect is:

$$\widehat{\text{ATE}} = 0.021 \quad \text{with 95\% CI: } [0.005, 0.037]$$

This estimate translates into an average increase in lifespan of approximately **1.57 years**, with a 95% confidence interval ranging from **0.36 to 2.79 years**.

Selected Covariates. The final causal forest model identified a set of variables that were most relevant for capturing treatment effect heterogeneity. These included several characteristics of the child, such as their age, the age of the oldest and youngest siblings in the household, and the total number of siblings (in particular, indicators for families with three to eight children). Maternal characteristics were also selected, including whether the mother was

divorced or never married, whether she was abandoned or her husband was institutionalized, and whether her marital status was missing in the records.

In addition, the model highlighted state-level policy and socioeconomic factors, including the ratio of state manufacturing earnings to national earnings, the statutory minimum age for child labor, the required age of school entry, the existence of continuation school requirements, and provisions stipulating whether mothers were obliged to work or to reapply for assistance. Measures of public spending were also prominent, such as total expenditures on social programs, state spending on charities, and school expenditures, as well as statutory allowances for the first and additional children. Finally, the model retained indicators for birth cohorts corresponding to the years 1909, 1915, and 1916, which may capture cohort-specific differences in conditions or program implementation.

A full list of all selected covariates is provided in Appendix Table 4.

5.2 Treatment Effect Heterogeneity

5.2.1 Omnibus Test for Treatment Effect Heterogeneity

To assess whether there is meaningful heterogeneity in treatment effects across individuals, I conduct an *omnibus test*, which provides a global check on whether treatment effects vary systematically in the sample, without linking that variation to specific covariates.

I split the trimmed sample into two subgroups based on the median estimated CATE: one group with CATEs above the median (high-effect group), and another with CATEs below the median (low-effect group). I then estimate the ATE separately for both groups and compare the results.

The findings indicate a notable difference in ATEs between the two subgroups:

$$\widehat{\text{ATE}}_{\text{High}} - \widehat{\text{ATE}}_{\text{Low}} = -0.054 \pm 0.022,$$

which corresponds to the 95% confidence interval for the difference in ATE. This supports the presence of statistically significant heterogeneity in treatment effects and confirms that the causal forest meaningfully captures variation beyond the average effect.

5.2.2 Formal Tests for Heterogeneity Along Key Covariates

To examine which covariates drive the observed heterogeneity in treatment effects, I conduct a series of formal hypothesis tests for each variable selected by the causal forest. The idea is to test whether the average treatment effect significantly differs between units with high and low values of a given covariate (split at the median), accounting for the clustered data structure.

Specifically, I compute a doubly-robust score for each unit, following the standard influence function formula, and aggregate this score at the cluster level. Then, for each covariate, I split the clusters into two groups based on the median of their mean covariate value. I perform a two-sample t-test comparing the mean doubly-robust scores between these two groups. This allows testing whether treatment effects are systematically different for high vs. low values of the covariate.

The results, reported in Appendix 4, indicate statistically significant heterogeneity along two dimensions:

- **Expenditures on schools**: Boys whose mothers lived in states with higher public spending on education experienced significantly *lower* treatment effects. The estimated difference is -0.064, with a p-value of 0.017.
- **State money to charities**: Similarly, boys living in states with higher levels of government support to charitable organizations also showed significantly lower treatment effects, with an estimated difference of -0.055 and a p-value of 0.040.

All other covariates exhibit non-significant differences in treatment effects, suggesting that observable heterogeneity is concentrated in a few meaningful predictors.

5.2.3 County-wise Heterogeneity of Treatment Effects

I further explored the heterogeneity of treatment effects at the county level. Figure 3 displays the distribution of estimated treatment effects across counties. Each bar represents the average treatment effect calculated within one county cluster.

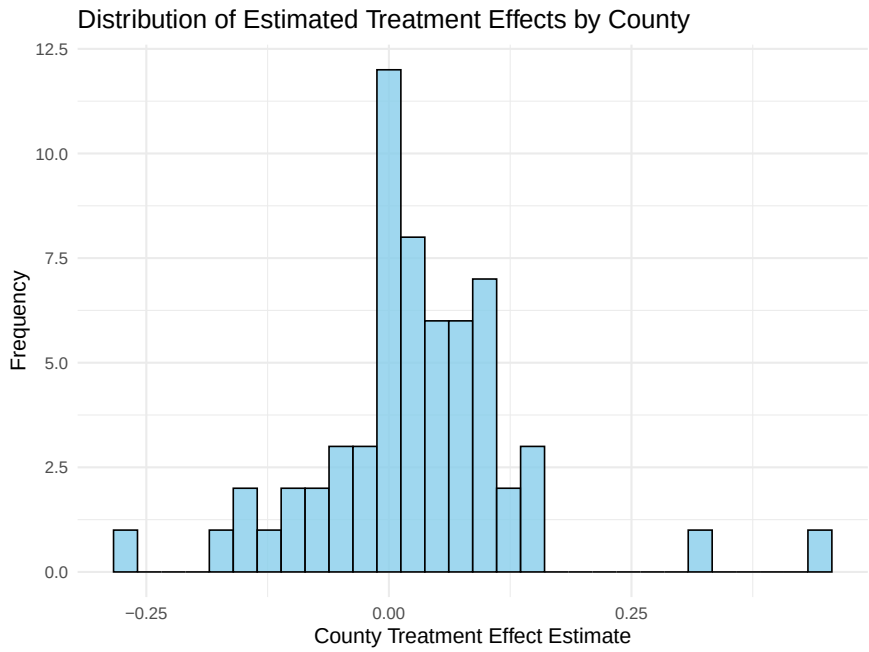

Figure 3: Distribution of Estimated Treatment Effects by County. Source: data from Aizer et al. (2016); author's own calculations.

The histogram shows that treatment effects vary substantially between counties. While the majority of counties cluster around an estimated effect close to zero, there is a notable spread, with some counties showing clearly positive average treatment effects, and others displaying negative estimates. This pattern suggests that the effectiveness of the program was not uniform across geographic areas. In particular, certain counties benefited more from the intervention, while others experienced limited or even adverse estimated impacts. This underscores the importance of accounting for spatial heterogeneity when interpreting overall program effects.

5.3 Robustness and Model Comparisons

5.3.1 Sanity Check: Re-estimating the ATE by County Means

To validate my main estimates, I computed a simple sanity check by averaging the estimated treatment effects across counties. Specifically, I calculated the mean and standard error of the county-level treatment effects and constructed the corresponding confidence interval. I then transformed these estimates back into the scale of years of lifespan to facilitate interpretation.

Both approaches — the county-level mean and the causal forest ATE — yield consistent evidence of a positive effect of program participation on longevity. The county-level mean

suggests an average increase of approximately 2 years, while the causal forest indicates a more precise estimate of 1.55 years. The confidence interval of the causal forest excludes zero, confirming statistical significance. The similarity of these results provides reassurance that the main findings are robust to alternative estimation strategies.

A summary of these comparisons is presented in Table 2.

Table 2: Comparison of ATE Estimates (Sanity Check vs. Causal Forest)

Method	ATE (log)	95% CI (log)	ATE (years)	95% CI (years)
County-level mean (sanity check)	0.03	[0; 0.054]	2.00	[0.03; 4.03]
Causal Forest ATE	0.02	[0.005; 0.037]	1.55	[0.35; 2.76]

Source: Data from Aizer et al. (2016); author's own calculations.

5.3.2 Variation in Propensity Scores Across Covariates

To explore whether the estimated probability of treatment assignment systematically varied across observable characteristics, I examined the distribution of propensity scores across several key covariates. This analysis helps assess whether there is evidence of systematic selection into treatment based on specific dimensions of the data.

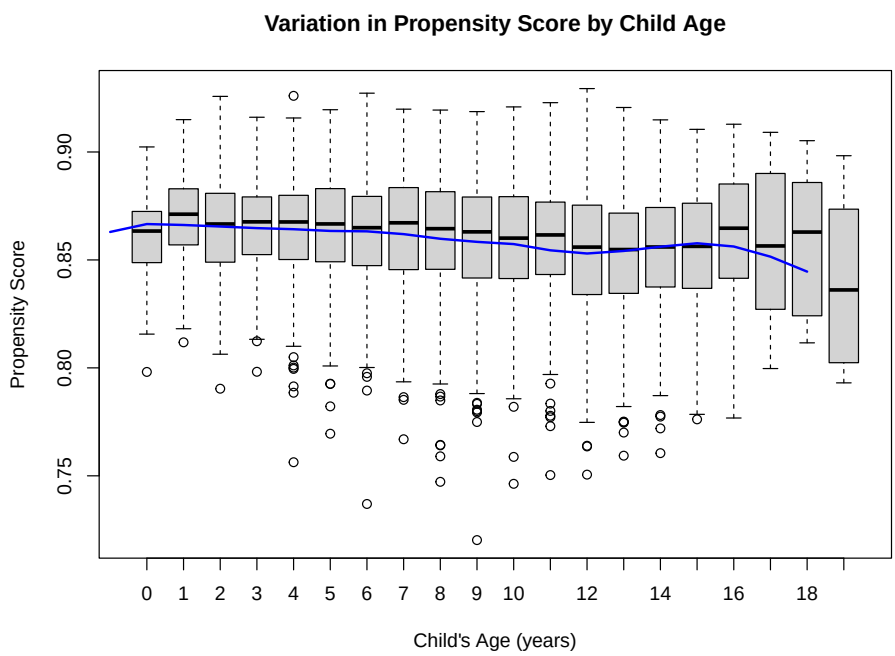

Figure 4: Variation in Propensity Score by Child Age. Source: data from Aizer et al. (2016); author's own calculations.

First, I analyzed how the propensity scores vary with the child's age. As shown in Figure 4,

the estimated probability of treatment remained relatively stable for younger children, with median scores around 0.85 to 0.88 across ages 0–13. For older children, particularly those aged 14 to 18, there is a modest decline in the median propensity score, suggesting that families with older children were somewhat less likely to have their applications accepted. Despite this downward trend, the interquartile ranges for each age group largely overlap, indicating that no severe confounding by child age is apparent.

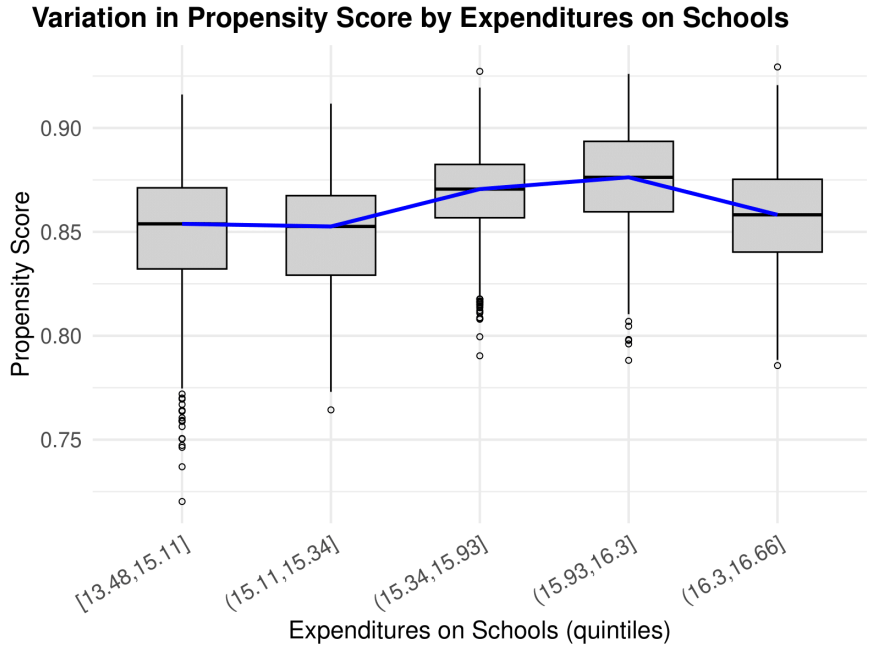

Figure 5: Variation in Propensity Score by Expenditures on Schools. Source: data from Aizer et al. (2016); author's own calculations.

Second, I considered the relationship between school expenditures and the estimated propensity scores. To do this, I grouped observations by quintiles of county-level expenditures on schools. The results, depicted in Figure 5, show a slight tendency for counties with higher education spending to have marginally higher propensity scores. However, the differences across quintiles are modest in magnitude, and the distributions remain broadly overlapping. This pattern suggests that while educational resources may correlate somewhat with treatment probability, this factor alone does not drive selection in a dominant way.

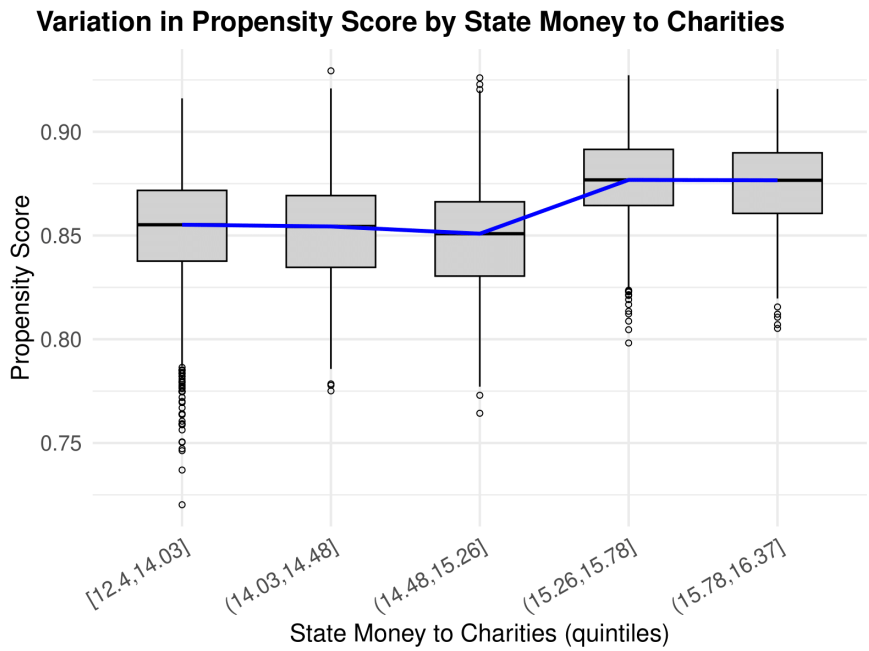

Figure 6: Variation in Propensity Score by State Money to Charities. Source: data from Aizer et al. (2016); author's own calculations.

Third, I examined the variation in propensity scores across quintiles of state-level expenditures on charitable programs. Figure 6 shows some evidence of a mild positive association, as counties in states with higher charitable spending exhibit slightly elevated median propensity scores. Nevertheless, as with school expenditures, the overall degree of overlap between quintiles is substantial, and no clear threshold effect is visible.

Taken together, these patterns indicate that while certain county- and state-level variables show small associations with treatment probability, the variation is not so pronounced as to imply severe imbalance across these dimensions. The trimming procedure and inclusion of these covariates in the causal forest specifications should therefore mitigate the potential for confounding related to these factors.

5.3.3 Analysis Ignoring Clustering

As a robustness check, I estimated the causal forest ignoring the clustering of observations within counties. While the main specification accounts for potential correlation in outcomes and treatment assignment within counties (by clustering), omitting clusters provides insight into how much clustering affects estimation precision and the heterogeneity detected by the model.

I first re-estimated the causal forest using the same set of selected covariates but without

supplying any cluster identifiers. The estimated Average Treatment Effect in this specification was slightly higher than in the main model:

$$\widehat{\text{ATE}} = 0.022 \pm 0.014,$$

indicating an average increase in log lifespan comparable to the model with clusters. However, the standard error is smaller, suggesting that ignoring within-county correlation tends to overstate precision.

To check whether the model without clusters was well-calibrated, I applied the calibration test (using crossfold predictions to avoid overfitting). This test regresses observed outcomes on the predicted CATEs and their mean prediction. Ideally, if the model is well-calibrated, the coefficient on the mean prediction should be close to 1, and the coefficient on the differential prediction (the heterogeneity component) should also be close to 1. The results showed:

- The coefficient on the mean prediction is 0.945 (significant at the 1% level), suggesting the model accurately captures the average treatment effect.
- The coefficient on the differential prediction is approximately -1.058 (not significant), implying that the model's estimates of heterogeneity across individuals are not statistically distinguishable from noise in this specification.

In other words, without clustering, the model correctly recovers the average effect but appears to overstate the precision of individual-level heterogeneity.

I further examined the distribution of estimated treatment effects by county. Figure 7 shows boxplots of CATE estimates for each county, ordered by their median treatment effect. The variation across counties is modest, with most medians clustered around 0.02, and no counties showing strong deviations.

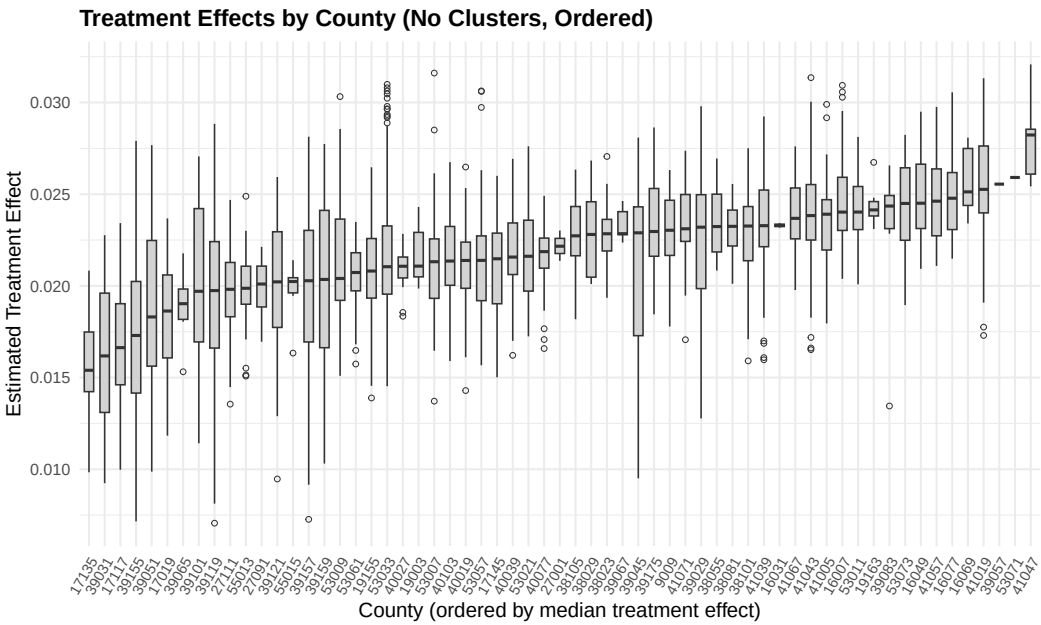

Figure 7: Estimated Treatment Effects by County (No Clusters, Ordered by Median). Source: data from Aizer et al. (2016); author's own calculations.

To further validate the model, I compared predictive performance using the R-loss metric (the residual squared error from predicting the transformed outcome). Compared to the main model with clusters, the non-clustered forest achieved a slightly lower R-loss, but the difference was extremely small:

$$R\text{-loss (no clusters)} - R\text{-loss (clusters)} = 1.1 \times 10^{-5},$$

$$R\text{-loss (crossfold)} - R\text{-loss (clusters)} = 3.0 \times 10^{-5},$$

confirming that ignoring clustering only marginally improved in-sample fit, likely by overstating degrees of freedom.

Finally, I performed an ANOVA of the doubly-robust scores on county fixed effects to test whether any residual heterogeneity was strongly related to counties. ANOVA partitions the total variance in the scores into within-county and between-county components and compares them to assess whether the means differ systematically across counties. The analysis yielded an F statistic of 0.8 with a p-value of 0.867, suggesting no statistically significant differences in the residual treatment effects across counties after adjusting for covariates and ignoring clusters.

Overall, this analysis confirms that while the point estimates remain similar when ignoring clusters, the uncertainty in heterogeneity estimates is understated. The main model's inclusion

of clustering therefore provides a more conservative and likely more accurate reflection of uncertainty.

5.3.4 Analysis Without Orthogonalization

In the final robustness check, I estimated a causal forest without orthogonalization—that is, without modeling the propensity score. Specifically, I set the estimated treatment probability W_i equal to the overall sample mean, thereby ignoring any adjustment for observed confounding.

The estimated average treatment effect in this specification was larger in magnitude:

$$\widehat{\text{ATE}} = 0.027 \pm 0.027.$$

However, the confidence interval included zero, implying the effect was not statistically significant at conventional levels.

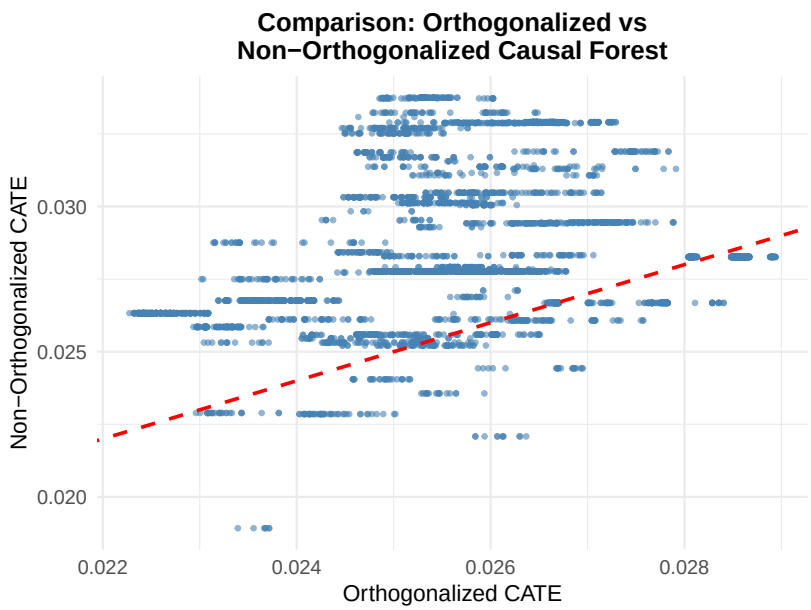

Figure 8: Comparison of Orthogonalized vs. Non-Orthogonalized CATE Estimates. The dashed red line denotes the 45-degree line. Source: data from Aizer et al. (2016); author's own calculations.

Figure 8 compares the Conditional Average Treatment Effects from the orthogonalized forest to those from the non-orthogonalized forest. The red dashed line shows the $45°$ line where the estimates would coincide. While the two sets of estimates are positively correlated, they diverge notably for higher predicted CATEs, suggesting that omitting orthogonalization

leads to bias or instability in treatment effect estimation. This finding highlights the importance of orthogonalization to correct for residual confounding and improve the robustness of the estimates. In this application, orthogonalization yields more conservative and credible estimates of both ATE and CATE.

6 Conclusion

This paper has examined the long-run effects of participation in the Mothers' Pension program on male children's longevity using causal machine learning methods. Employing causal forests, I estimated both average and heterogeneous treatment effects in a rich dataset linking early 20th-century administrative welfare records to mortality outcomes.

The analysis produced several consistent findings. First, in multiple specifications, including models with and without county-fixed effects and after trimming observations with extreme propensity scores, the estimated average treatment effect was positive, indicating that receiving MP support increased lifespan by approximately 1.5 to 2 years on average. The estimated effect was substantial, and its statistical significance in the preferred specification highlights the potential of targeted social transfers to generate long-term meaningful improvements in life outcomes.

Second, examining treatment effect heterogeneity revealed that the benefits of the program were not uniform. An omnibus test confirmed that estimated CATEs varied systematically across individuals. Specifically, children from states with lower public expenditures on education and charities appeared to derive somewhat larger gains in longevity, although most other covariate-level differences were not significant. These findings indicate that the program's impact was more pronounced in areas with limited public resources. This may reflect that, in such areas, families started from a lower baseline of welfare and had fewer alternative means of support, so each transfer potentially produced a larger marginal improvement in children's long-term success.

Third, several robustness checks—including comparisons to simpler county-level means, models estimated without orthogonalization, and models estimated without clustering — highlighted the importance of appropriate adjustment for confounding and clustering. The orthogonalized causal forest estimates were more stable and conservative, underscoring the relevance of modern machine learning tools for causal inference in historical data settings.

Accounting for clustering ensures that standard errors and confidence intervals accurately reflect the dependence of observations within counties, providing a more conservative and realistic assessment of statistical significance.

More broadly, this research illustrates how combining rich administrative records with modern nonparametric methods can uncover long-term impacts of early social policies. While the estimated effects are broadly consistent in magnitude with previous research, they remain meaningful in both statistical and substantive terms, given the limited size of the transfers and the historical context of the intervention.

Future work could further explore heterogeneous effects by leveraging additional covariates, refining matching strategies, or extending the analysis to female children. Overall, the results provide new evidence that social assistance to vulnerable families during childhood can produce sustained improvements in lifespan.

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Table 3: Summary Statistics for Estimation Sample (Matched Data)

Variable	Mean.Treated	Mean.Control	Difference	SE..Diff.
Child age (years)	8.3082740	8.8270600	-0.5187861	0.1435639
Length of family name	6.3603315	6.3448627	0.0154689	0.0555755
Number of kids in family (imputed)	3.7306965	3.5381485	0.1925480	0.0674464
Age of oldest kid in family record	11.4656777	11.8606307	-0.3949530	0.1532161
Age of youngest kid in family record	4.8551483	5.6297050	-0.7745566	0.1330047
Divorced or never married	0.0306820	0.0295015	0.0011805	0.0057863
Husband abandoned, in prison/hospital	0.1877272	0.1709054	0.0168218	0.0129023
Mother's marital status unknown	0.2326596	0.2675483	-0.0348887	0.0150174
Day or month of birth missing	0.0183219	0.0071211	0.0112009	0.0031330
Mean Duncan socio-economic Index	24.9602532	24.3798555	0.5803977	0.1114874
SD of Socioeconomic Index (County)	20.7241651	20.8169928	-0.0928276	0.0784646
% living in urban areas	0.4237997	0.3594838	0.0643158	0.0085991
% females that are widowed	0.0631971	0.0613168	0.0018803	0.0007746
% under 16 with mom's widowed, divorced single, abandoned	0.0550207	0.0558320	-0.0008113	0.0010640
% workers with SEI¡20th percentile	0.1014681	0.1100010	-0.0085330	0.0013697
% women 16+ in the labor force	0.1825097	0.1816136	0.0008961	0.0014467
% children 10-15 working	0.0709636	0.0661017	0.0048620	0.0018335
Average value of farm land	94.9315108	80.8362157	14.0952952	1.9907803
Ratio of state manufacturing annual earnings to U.S. manufacturing earnings	1.1034770	1.1191719	-0.0156950	0.0029816
Age of School Entry, CG	7.6162571	7.5727365	0.0435206	0.0321915
Min Age Child Labor, CG, 0=none	14.5839756	14.7863683	-0.2023927	0.0290286
Continuation Schl Req = 1, CG	0.3696379	0.2451679	0.1244701	0.0149111
Total expenditures on social programs	13.5983945	13.6026844	-0.0042900	0.0460934
State money to charities	14.8929956	14.9192696	-0.0262740	0.0356037
Expenditures on schools	15.6099137	15.5015424	0.1083713	0.0248617
MP law requires receipient to work	0.5605642	0.6775178	-0.1169536	0.0160722
MP law requires re-application	0.4840774	0.5737538	-0.0896765	0.0168927
Amount for first child in state law	17.9492511	19.5350966	-1.5858455	0.3272513
Monthly allowance for second child	7.9631380	8.0930824	-0.1299444	0.0945130
YOB of child	1913.5928457	1912.1627670	1.4300787	0.2240318
Log of Age at Death (MP Birth Date)	4.2826655	4.2694595	0.0132060	0.0057249

Source: Data from Aizer et al. (2016); author's own calculations.

A Appendix

Table 4: Heterogeneity of Treatment Effects across Covariates.

Covariate	Mean (High)	Mean (Low)	Difference	95% CI lower	95% CI upper	p-value
Expenditures on schools	-0.005	0.059	-0.064	-0.116	-0.012	0.017 *
State money to charities	-0.001	0.055	-0.055	-0.108	-0.003	0.040 *
Total expenditures on social programs	0.007	0.047	-0.040	-0.094	0.014	0.139
Allowance for first child	0.003	0.038	-0.035	-0.082	0.012	0.141
Number of children: 8	0.047	0.009	0.038	-0.016	0.092	0.163
Min. age child labor (CG)	0.007	0.037	-0.030	-0.080	0.020	0.229
Age of school entry (CG)	0.011	0.043	-0.032	-0.086	0.022	0.241
Husband absent	0.011	0.043	-0.031	-0.086	0.023	0.251
Continuation school requirement (CG)	0.041	0.014	0.027	-0.027	0.082	0.318
State manufacturing earnings ratio	0.017	0.037	-0.019	-0.074	0.035	0.478
Marital status missing	0.037	0.018	0.019	-0.035	0.074	0.481
Length of family name	0.036	0.019	0.018	-0.037	0.072	0.513
Law requires recipient to work	0.020	0.035	-0.015	-0.069	0.040	0.587
Age of youngest child	0.033	0.021	0.012	-0.043	0.067	0.667
Law requires re-application	0.033	0.022	0.011	-0.044	0.067	0.687
Child age (years)	0.033	0.022	0.011	-0.045	0.066	0.706
Number of children: 5	0.022	0.032	-0.009	-0.064	0.045	0.731
Birth year: 1915	0.025	0.030	-0.005	-0.059	0.049	0.862
Age of oldest child	0.029	0.025	0.004	-0.051	0.059	0.894
Birth year: 1909	0.026	0.029	-0.003	-0.058	0.052	0.916
Birth year: 1916	0.026	0.029	-0.003	-0.058	0.052	0.919
Allowance for additional child	0.029	0.026	0.003	-0.052	0.058	0.922
Divorced or never married	0.029	0.027	0.002	-0.047	0.051	0.945
Number of children: 6	0.028	0.027	0.001	-0.053	0.056	0.957
Number of children: 3	0.028	0.027	0.001	-0.053	0.056	0.962
Number of children: 4	0.027	0.028	-0.001	-0.056	0.054	0.970

Note: This table reports the difference in estimated treatment effects between units with high and low values of each covariate (split at the median). p-values are reported with significance stars: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.
Source: Data from Aizer et al. (2016); author's own calculations.